



Advancements in wood processing utilizing polymers for “transparent” wood composites: a technical review and *Meta-analysis*

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Abstract

Concerns about environmental deterioration, resource depletion and climate change have fueled a surge in worldwide interest in sustainable materials these recent years. This interest is especially strong in businesses that rely on non-renewable resources, such as building, transportation and packaging. In this paper, there has been a process toward investigating alternative materials, which have both environmental advantages and functional capabilities equivalent to, or even superior to, traditional counterparts. This review and *meta-analysis* aims to assess the present status of technical research on “transparent” wood manufacture, by utilizing even recyclable polymers. This study evaluates the methods used, the attributes gained, obstacles encountered and possible uses of such new materials. A technical literature search was undertaken utilizing several databases, such as PubMed and Web of Science and pertinent scholarly journals. The inclusion criteria included peer-reviewed studies on alternative wood production utilizing various polymers. The data extraction includes polymer types, production procedures, optical and mechanical properties and limitations described in the research. Several technologies, including impregnation and hot pressing, have been used to create novel composites. “Transparent” wood composites demonstrated promising optical transparency, mechanical strength and thermal stability when compared to standard approaches. Scalability, durability and cost-effectiveness have been referred as major problems in the manufacture of “transparent” wood composites, so to the moment their market impact is low. Despite limitations, the accurate research revealed potential uses in design, renewable energy and the sustainable packaging industries.

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Introduction

Known as “Transparent wood composites”, the subject has received particular attention in recent years, with the first igniting study in the area conducted by Fink in 1992. From 2001 to 2010, organic transparent materials, especially membranes made from lignocellulosic materials (mainly nanocellulose), were the focus of research. Some examples of scientific publications related to the topic of transparent materials based on cellulose, are those of manufacturing nanocellulosic membranes (Nogi et al. 2008; Moon et al. 2011; Zhu et al. 2016a, b; Abrial et al. 2020).

According to the literature, while there is increasing interest in “transparent” wood as a sustainable structural material, literature on the use of recyclable polymers in such composite materials in these cases, is relatively limited. There is a need for more studies to explore the use of different types of recyclable polymers applied, and their effects on the properties of the resulting composite materials. The types of polymers used in the impregnation of “transparent” wood composites vary, but common types include epoxy resins, polyesters and polyurethane resins (Hou et al. 2023). The crucial characteristic for the polymers chosen is their capability of initial polymerization and then subsequent cross-linking. These polymers have primarily and foremost great optical properties, then been found to improve the mechanical and thermal properties of wood composite materials. Epoxy resins are often colorless and used as adhesives for gluing glass surfaces, since they provide good transparency after curing. Epoxy resins are yielded by polycondensation reactions, after mixing certain volumes of two fluid parts; the oligomer and the hardener (Sideridou et al. 2016). Polyesters are a vast category of polycondensation polymers as well, with great range of qualities. Concerning vinyl products, poly(methyl methacrylate) is the major member of the acrylics family, providing a transparent, colorless and light-weight polymer. Acrylics are produced by radical polymerization, upon heat, with bulk or emulsion technique. PMMA is widely used in high-demanding applications due to its excellent optical properties, such as high light transmission and low scattering (*pe. plexiglas*) (Achilias 2007).

As for the impregnation techniques used, they vary depending on the desired properties of the final product and the viscosity of the monomer, oligomer or prepolymer induced. Common techniques include vacuum impregnation, pressure impregnation and infiltration with solvent gel. Those techniques have been found to improve the transparency, mechanical strength and thermal conductivity of the resulting composite materials. An emerging application field is the creation of cellulosic composite materials with polymers of very good mechanical properties, which were manufactured by the final densification of wood samples, previously delignified (Frey et al. 2018). Densification refers to the post-treatment of wood, mechanical or chemical, that provides denser wood specimens. The delignification process involves the removal of some, or all, the lignin from the wood or the modification of lignin to create an advanced strong and durable scaffold. At the same time improved dimensional stability in combination with densification or impregnation may take place to enhance mechanical/optical properties (Wang et al. 2021). This process can be achieved through various methods, such as alkaline or acidic delignification, as well as reductive catalytic fractionation (Song et al. 2018). An important factor to con-

sider, before the selection of the technique, is the degree of delignification to achieve, before introducing the polymer into the wood mass.

The main challenges in creating “transparent” wood are to increase the thickness (of the generally thin materials applied), improving light diffusion and flame retardancy, while adding functions such as resistance to extended exposure to UV and IR radiation or hydrophobia and humidity repellency (for final properties’ maintenance). According to the literature, the eventual properties of “transparent” wood composites impregnated with polymers depend on various factors, including the type of wood, polymer and impregnation technique applied. In general, it has been found that the addition of polymers increases the strength, durability and flexibility of the resulting composite materials. Furthermore, “transparency” of composite materials can also be improved with certain impregnation techniques and polymers. Transparency of a material is defined as the percentage of the light rays the material allows to penetrate its mass, without absorbing or scattering it (Gigilashvili 2022). The visual result of this physical property is that we may see through that transparent material well, as it is through the air (*pe.* glass is transparent). On the other hand, haziness is defined as the amount of light subjected to scattering, when penetrating a material, greater than 2.5 (divided by the total light transmission according to ASTM D1003 standard 2000). The visual result of this physical property is that we may see through that haze material a foggy or milky image, compared to the clear one (*pe.* plastic films are haze). Apart from homogenous materials, such as polymers, in composite materials the key-factor regarding transparency is the samples’ thickness (Guzman-Puyol 2022). This literature review highlights the potential of certain polymers in improving/altering the properties of “transparent” wood composite materials. However, more research is needed to explore the full potential of these industrial manufacturing processes and address current research gaps in the field, since many parameters are involved.

The objectives of this study were (1) to briefly analyze the selected review papers (*meta-analysis*) published since 1992, based on general interesting data (e.g., years, journal of publication, etc.) or details regarding environmental issues; (2) to review the polymers types used to impregnation of “transparent” wood, the production procedure like delignification (solvents, time, temperature, pressure) and impregnation (temperature, time, pressure); (3) to discuss applications of “transparent” wood composites; (4) to review the improved properties of “transparent” wood composites, such as optical transparency, mechanical strength, and thermal stability and (5) to briefly discuss the gaps in the literature, as well as potential future actions or goals. The importance of monitoring and the *meta-analysis* of the published review papers are also the innovation of this paper.

Materials, methods and bibliometric analysis

Research methodology

This review study followed the optimal reference criteria for systematic reviews and *meta-analyses* established by Moher et al. (2009). The first segment consists of systematic reviews. Unbiased summaries provide a comprehensive summary of current

research topics. The second section is a *meta-analysis*, which integrates results using various statistical approaches and aggregates data. The PRISMA technique aims to help researchers and practitioners arrange their literature review reports (Liberati et al. 2009). Previous review studies have used the PRISMA technique, according to the study for this publication. The three main PRISMA procedures were followed herein too [PRISMA flow Diagram]: identification, literature review, screening and selection of qualified publications based on keywords, eligibility and data export and then finalization.

Data collection/bibliometric analysis for the *Meta-analysis*

The Elsevier Scopus[®] and Google Scholar databases were chosen as a starting point for the search for pertinent material. The purpose of this choice was to look for publications in a very broad context, thus the literature search was conducted by looking for revising papers based on keywords, such as “transparent wood”, “recycled polymers”, “optical properties”, “delignification”, “mechanical properties”, “solid wood” or “plastics”, as well as combining keywords related to “transparent wood” and “delignification factors”. Considering the time periods, the search was broad enough from 1970 to 2023. After conducting the proper searches in the relevant databases, all of the results in Elsevier Scopus[®] and Google Scholar were exhausted. Case studies, book chapters, conference papers, postgraduate and doctoral theses along with papers written in a language other than English, were all eliminated from the search. Review papers were, therefore, screened by title, keywords and abstract, while keeping in mind the Inclusion/Exclusion criteria, so that the pertinent publications were added in our list. Each paper’s main body was examined, paying close attention to the abstract, methodology and conclusion sections in particular. The Data was extracted from selected studies, including variables such as polymer types, fabrication techniques, optical properties, mechanical properties, delignification methods.

The retrieved papers most frequently appeared the terms “transparent wood”, “delignification of wood”, as keywords, showing that throughout the previous ten years, research on the field was very active (Fig. 1a). Many of the journals that have published reviews of “transparent wood” are regarded as high-impact journals, like “Renewable And Sustainable Energy Reviews” (Fig. 1b). For the past years 10 years, scientists from all over the world have contributed to the advancement of knowledge in the field of study on “transparent” wood composites. In addition to several other smaller groups in other countries of the globe, the major spatial clusters of institutions/universities working on this topic are dispersed in 37 countries, throughout America, Europe and Asia. China has provided the most reviews overall, by far (Fig. 1c). The countries that follow are the USA, Canada, Germany, Finland, Sweden, UK, India, Italy, Japan, Australia, Austria, and France.

Bibliometric analysis of all articles for “transparent wood”

In this section, an analysis is carried out regarding all articles published concerning “transparent wood”. This very review publication aims to summarize the data and the results by using the same method by Moher et al. (2009), as in the first part.

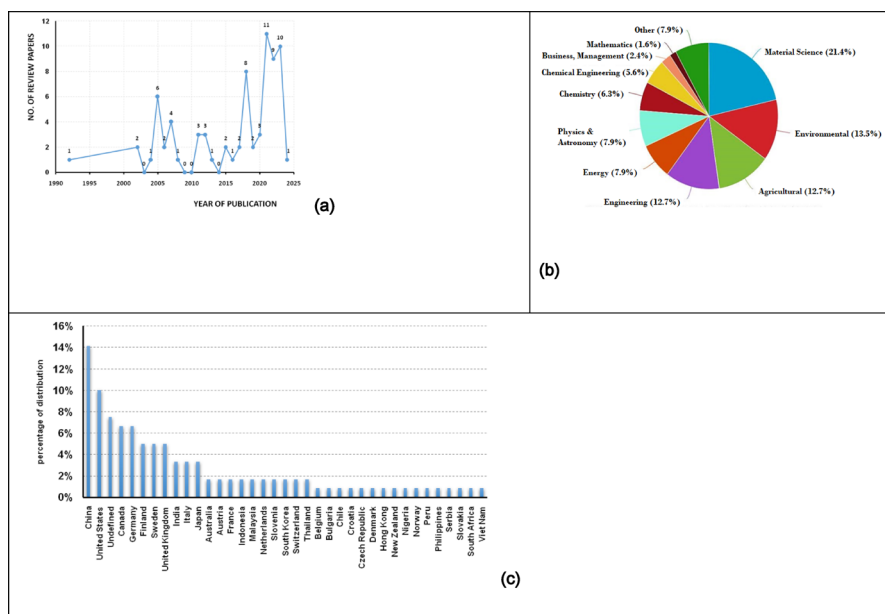


Fig. 1 Distribution of selected 57 review papers regarding “Transparent wood”, based on (a) year of publication, (b) subject area, while (c) presents the countries with the most review articles published on “Transparent wood”

The initial step of the search for applicable content was launched from the Scopus[®] and Google Scholar databases. This selection was made with a fairly broad context in mind, and the literature search was conducted by looking for papers that contained studies on “transparent wood composites”. The study used keywords such as “transparent wood”, “polymers”, “delignification” and “impregnation”, as well as a combination of “transparent wood”- and “delignification”-related phrases. The search was restricted to items from 1992 to 2024 due to the time frames. After doing the appropriate searches in the pertinent databases, all Scopus[®] and Google Scholar results were exhausted, and then stopped when there were several pages of results with no relevant publications, as we did with the *meta*-analysis. The following were excluded from the search: book chapters, conference papers, master degree and doctoral theses, while anything written in a language other than English too. Considering the criteria, the papers were subsequently checked by title, keywords and abstract, so the relevant publications were noted in our list. The main body of each document was studied, paying particular attention to the Abstract, Material & Methods and Conclusions sections. The obtained papers were then put through a bibliometric analysis and Fig. 2 shows the results of the analysis; the quantity of pertinent papers is increasing yearly.

Among the 701 documents acquired, 57 were examined and further categorized. There were a significant number of studies since 2017 that discussed “transparent” wood composites, with main keywords “Wood”, “Cellulose”, “Transparent”, “Tensile Strength”. Studies on “transparent wood” are published in numerous scientific journals, many of which are recognized as having high impact factors. The journals “Cel-

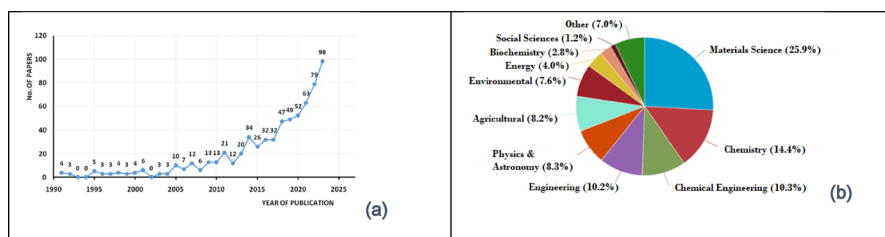


Fig. 2 Post-treated results of literature research: **(a)** Distribution of review papers based on year of publication. **(b)** Number of relevant review papers by subject area

Table 1 The countries with the most relevant publications on “transparent wood composites”

Country	Percentage of publications
China	25.27%
United States	11.93%
Sweden	6.51%
Japan, Canada, Finland, Germany, Slovakia, France, Italy	3.58%
India	2.28%
Czech Republic, Switzerland, United Kingdom, Spain, S. Korea, Austria, Slovenia, Belgium	2.06%
Hong Kong, Malaysia, Netherlands	0.98%
Australia, Denmark, Iran, N. Zealand, Russian Federation, Saudi Arabia, Mexico, Romania, Norway, Poland, Brazil, Croatia, Egypt, Thailand, Turkey	>0.87%

lulose”, “Progress In Organic Coatings”, “ACS Applied Materials And Interfaces”, “Coatings”, “ACS Sustainable Chemistry And Engineering”, “Industrial Crops And Products”, “Journal Of Applied Polymer Science”, “Bioresources”, “Chemical Engineering Journal”, “Holzforschung”, “Polymers”, “Carbohydrate Polymers”, “Bio-macromolecules”, “European Journal of Wood And Wood Products” are among them during the past few years.

Over the past few years, researchers studying “transparent” wood from all over the world have advanced our knowledge in this area. The largest spatial clusters of institutions researching on this topic are scattered throughout 60 countries, including the America, Europe and Asia, in addition to numerous additional smaller groups in other parts of the world. China has, by far, contributed the most overall. According to Table 1, the countries that participated in the chosen publications, China, USA, Sweden, Japan, Canada, Finland, Germany, Slovakia, France and Italy had the most articles published in the previous years. Once more, as with the results of the *meta*-analysis previously, these studies show that these areas place a high emphasis on addressing the need for new sustainable material because they were carried out in both economically-developed and less economically-developed countries, as well as in densely populated areas. The countries with higher publication rates typically have active scientific communities and do invest in research and innovation in the specific field. This can be explained by several means, such as research funding, the presence

of specialized research centers, scientific and educational institutions, the abundance of raw wood material perhaps, as well as the prior experience and expertise of scientists working in these countries [The Role of Wood Science and Technology in a Sustainable World 2018].

Results

To begin with the technical description of this chemical and engineering process, to produce “transparent” composite materials from wood, the following steps need to be taken (Fink 1992; Li et al. 2016a; Montanari et al. 2020; Mi et al. 2020a, b): (i) Firstly, lignin, chromophores, colored tannins and other polyphenols need to be removed, known as delignification process. The wood produced tends to have a less yellow hue, and then thin pieces often appear “transparent”. (ii) Secondly, the pores/spaces in cell walls created due to the removal of lignin, tend to increase the specific surface area without affecting the anisotropic structure of the wood, and can subsequently be filled using a polymeric matrix with the same refractive index. After polymerization, the matrix is cross-linked with the lignin left; this process is called setting or curing (Fink 1992). After that, as understood, the product manufactured is a composite material including wooden scaffold and polymeric filler, surely cross-linked and amorphous (thus, “transparent” composite is correct over “transparent wood” term). In 1992, Fink pioneered the concept of “transparent” wood composites. Let us review some valuable recent contributions in the field since then.

In Shams et al. (2012) created a transparent material from the shells of crabs, from which they removed inorganic CaCO_3 particles, proteins, lipids and pigments. The remaining nanostructured chitin (chitin is the main polymer of shells, the second most abundant polysaccharide in nature behind cellulose) was impregnated with a monomer (CHEBI:60027) and polymerized, creating a category of materials suitable for transparent substrates in next-generation electronic devices, such as flexible screens and solar panels.

In 2016, efforts began to create transparent multifunctional structural materials. Zhu et al. (2016a, b), created anisotropic, highly “transparent” composite wood. NaOH (>98%), Na_2SO_3 (>98%) and H_2O_2 (30%) were used for lignin removal from Basswood (*Tilia Americana*). The polymer for impregnation was an epoxy resin (epoxy resin set). The solvents used were EtOH and deionized H_2O . The lignin removal solution consisted of dissolved NaOH (2.5 mol/L) and Na_2SO_3 (0.4 mol/L) in deionized H_2O . Samples of 50 mm × 50 mm and thickness 3 mm were immersed in this solution, boiled for 12 h and then rinsed in hot distilled H_2O . Then, samples were subjected to a bleaching solution (H_2O_2 , 2.5 mol/L). After the yellow color disappeared, the samples were rinsed with cold H_2O and the lignin-extracted samples were preserved in EtOH. The epoxy resin was prepared by mixing two liquid components (epoxy resin set) in a 2:1 ratio, as indicated. The wood, deprived of lignin, was immersed in the viscous resin under vacuum to get it into the wood structure. This process was repeated three times and the plate containing the wood and resin was kept at 30 °C for 12 h. They compared the filling directions with aligned channels, either vertically (*R*-wood) or longitudinally (*L*-wood) to the wood plane.

The research delves into the anisotropic optical and mechanical properties of these composite materials. “Transparent” wood composite type *L* exhibits higher strength, coefficient and ductility than “transparent” wood composite type *R*. The examination of the cross-section after stress-strain tests shows differences in fracture surfaces, with *L*-composite breaking inside the channels and *R*-composite breaking between the channels. It is known that mechanical forces apply in certain directions and geometries for evaluating the mechanical profile of a material/specimen.

In Li et al. (2016a, b) presented the construction and characterization of optically “transparent” wood composites with high transmittance (85%) and haze (71%). They used wood of the species *Ochroma pyramidale*, (balsa wood) with dimensions of 20 mm × 20 mm and various thicknesses of 0.6, 1.0, 2.5, 5.0 and 8.0 mm. Delignification was performed with 1% NaClO₂ with an (COOH)₂ buffer solution (pH 4.6) at 80 °C. The reaction time was 6 h for samples less than 3 mm thick and 12 h for samples 5 and 8 mm thick. For impregnation, pure monomer methyl methacrylate (MMA) was used, which was pre-polymerized at 75 °C for 15 min in a round-bottom flask with 0.3% *N*, *N*′-azoisobutyl nitrile (AIBN) as a free-radical initiator. The pre-polymerized MMA was cooled to room temperature in an ice water bath to terminate the reaction, when its viscosity was the proper one for penetration. The delignified wood template was immersed in the pre-polymerized MMA solution under vacuum for 30 min, with three repetitions for complete impregnation. The impregnated wood was “sandwiched” between two glass plates, wrapped in aluminum foil and polymerized by heating in an oven at 70 °C for 4 h. Synergy between wood and poly(methyl methacrylate) (PMMA) was studied, regarding mechanical properties.

Zhu et al. in (2016) created “transparent” wood composite for application as a material for solar panels. They used basswood as well, which was delignified in a solution containing NaOH and Na₂SO₃ in deionised H₂O by boiling for 12 h, followed by rinsing in hot distilled H₂O and subsequent treatment in a bleaching solution (H₂O₂). The samples delignified were preserved in EtOH. Polyvinylpyrrolidone (PVP) was dissolved in EtOH too, in order to form a solution with a concentration of 15%. The wood was immersed in the PVP solution, under vacuum (200 Pa) for about 10 min, ensuring complete penetration. Finally, the wood piece was placed on a hot plate at 60 °C. The cured wood sample was detached after complete evaporation of the solvent. A drop of ethanol was deposited on the solar cell for collision. The “transparent” wood composite was placed on the cell to create close contact. The multilayer was allowed to dry at room temperature until the wood was firmly attached. “Transparent” wood composite exhibited high optical transmittance (90–75%) from Vis to near-IR wavelengths (400–1100 nm). The transparency of a 1 mm-thick wood composite is comparable to transparent glass, plastic and cellulose-based nanopaper. The transparency is the result of successive lignin removal and PVP infilling processes. The wood sample exhibits autofluorescence when excited at UV rays. This could be attributed to the remaining presence of lignin and its aromatic structures that contain conjugated bonds. Microscopic analyses confirm the unique microstructure of wood microchannels and fibers. Microfiber units contribute to light reflection/scattering and microchannels do guide incident light for forward scattering. “Transparent” wood composite materials exhibit high haze transmittance (80%) from 400 to 1100 nm. High haze is attributed to the microstructure of wood fibers and channels, which pro-

vide effective light reflection/scattering (Fang and Zhu 2014). “Transparent” wood composite can serve as an effective cover or light management substrate for optoelectronic devices. “Transparent” composite serves as a substrate for a GaAs solar cell, demonstrating increased short-circuit density and overall conversion efficiency. The wood layer acts as an anti-reflective level and light trapping layer, improving the paths of photons in the active layer of the solar cell. Enhanced performance is attributed to a combination of forward scattering and refractive index matching.

Li et al. (2016a, b) developed a transparent material from wood aiming to replace glass. They used Basswood, which was immersed in a solution containing NaOH and Na_2SO_3 for 3 h under boiling conditions. This was followed by immersion in boiling H_2O_2 until the wood became white. The white wood was then immersed in a liquid epoxy resin for complete penetration. “Transparent” wood composite was obtained after the epoxy resin solidified completely. This new material exhibited transparency exceeding 85% in the visible wavelength range and greater than 95% haze, ensuring uniform and consistent distribution of daylight. The scattering ratio was 9 for a 0.5 cm thick “transparent” wood composite, providing a unique light guiding phenomenon. Thermal conductivity was approximately 0.32 W/m·K, along the wood growth direction, and 0.15 W/m·K in the transverse plane, significantly lower than glass of course, which could replace in terms of transparency, at certain occasions (for glass it is approximately 1 W/m·K). Enhanced energy absorption capabilities were achieved, regarding impact, eliminating safety issues associated with glass.

In Li et al. (2017a, b) aimed to create a “Phosphorescent Transparent Wood”, since functional light-conversion layers containing active phosphorescent materials are essential components in many devices, like white light-emitting diodes (LEDs). In LEDs blue emission from a GaN-based diode is partially converted to red and green emission using a phosphor layer. The researchers examined the construction and properties of “transparent” wood composite materials, focusing specifically on the incorporation of phosphorescent quantum dots (QDs) into the material. Wood delignification was achieved using NaOH, Na_2SO_3 and H_2O_2 . The resulting wood was immersed in liquid epoxy resin. Optical properties such as haze, forward transmittance and backward reflectance were measured. The “transparent” wood composite exhibited high optical transparency (>85%) and optical haze (>95%). The use of phosphorescent QDs (Si and CdSe) in monomer MMA led to phosphorescent wood composite. Phosphorescence showed diffuse red and green light under excitation with blue/UV light. Scattering losses were evaluated, with Si QDs exhibiting negligible reabsorption at the emission wavelength. Light scattering power depended on the direction of the wood fibers, maintaining the wood’s original structure. Monomer MMA, enriched with phosphorescent QDs, was pre-polymerized before penetrating the delignified wood, for viscosity increase.

Gan et al. (2017a), created a new phosphorescent and “transparent” composite material from wood. They delignified Linden wood (20 mm × 10 mm × 0.5 mm) using a solution of NaClO_2 (2% w/w) and $(\text{COOH})_2$ solution (pH=4.6) at 80 °C for 12 h, followed by immersion in a H_2O_2 solution (5 mol/L) and boiling for 4 h. Subsequently, it was dried at 45 °C for 24 h. MMA and AIBN were mixed in a weight ratio of 1000:3. The mixture was pre-polymerized at 75 °C for 20 min. The pre-polymerized MMA solution was cooled to room temperature using an ice water bath,

later the solution and $\gamma\text{-Fe}_2\text{O}_3@\text{YVO}_4:\text{Eu}^{3+}$ nanoparticles were uniformly dispersed and mixed. Different weight ratios (1000:1, 1000:2, 1000:5 and 1000:10) were used to achieve “phosphorescent” wood composite with different nanoparticle concentrations. The delignified wood scaffold was impregnated with the mixture under vacuum conditions for 30 min, with three repetitions to ensure complete impregnation. The impregnated wood was placed between two glass substrate plates at 50 °C for 6 h. The wood composite exhibited high optical transparency over Vis wavelength range (350 to 800 nm). The “phosphorescent” wood composite exhibited intense colored photoluminescence under excitation with UV radiation at 254 nm.

Researchers cited in Wu et al. (2019) created “transparent” wood composite by applying 5 different delignification times and impregnating pre-polymerized MMA mixed solution upon heating. Surface color, optical transmission and optical haze, along with mechanical properties were measured. “Transparent” wood exhibited maximum optical transmittance of 61% at 800 nm (Vis-range border) with a lignin content of 9%, whereas composites had enhanced optical haze at different delignification levels. Furthermore, “transparent” wood composite has shown maximum tensile strength of up to 171.4 MPa with 15% lignin content. The innovation of this method lies in the fact that the manufactured “transparent” wood composite retains the natural color and texture of wood while emitting light.

Researchers found in Bi et al. (2018) utilized deep eutectic solvents for preparing “transparent” wood composites for LED applications. In the wooden substrates, they impregnated acrylic acid diluted in deionized H_2O at a concentration of 40% w/w in the presence of superoxide ammonium, followed by pressurized impregnation at 200 Pa (for ≈ 10 min) to ensure complete penetration of the solution into the wood, followed by heating (at 75 °C for 4 h).

Epoxy resins and PMMA are the often-used polymers, as seen so far, in the impregnation stage of delignified wood specimens. However, some other polymers have also been found for these processes in the literature. For example, Rao et al. (2019) constructed “transparent” birch wood (*Betula*) using poly(vinyl acetate) (PVA). The obtained “transparent” wood composite samples were highly flexible, and the optical and mechanical properties of the composite wood-PVA samples were significantly improved by adding poly(ethylene glycol) (PEG) as a plasticizer. The wood-PVA composites, with or without PEG, showed high flexibility and brittleness. The optical transmission of those samples was at 65% and 69%, respectively, while the haze of the samples was 80% and 85%, respectively. Once increasing PEG content in PVA to 100 wt%, the optical transmission and haze increased to 80% and 90%, correspondingly. Thus, this case of composite materials exhibits potential applications in diffuse light shaping, decorative lighting, optoelectronics or displays.

In the work of Mi et al. (2020a, b), aesthetical “transparent” wood was developed by removing lignin and impregnating it with epoxy resin, not only to manufacture the “transparent” composite, but also to retain its natural patterns simultaneously. The material exhibited excellent optical transparency, ultraviolet exclusion, thermal insulation, mechanical strength and aesthetics. Such materials must yield out uniform illumination and require high mechanical and hydrophobic properties. However, they are rarely environmentally friendly. To address this issue, the research team of Fu et al. (2020a, b) created a biologically-based film (polymer matrix-free in this case)

with photoluminescence and hydrophobicity for use in optical lighting. They incorporated an organic solution of CdSe/ZnS QDs, modifying the microstructure of the delignified wood through chemical processing. Then, they densified it and coated it with a thin layer of hydrophobic hexadecyltrimethoxysilane through chemical vapor deposition, creating eventually a composite material with potential optical applications, such as solar panels and laser devices (Fu et al. 2020a, b).

In 2017–2018, the research community directed its focus towards the development of multi-layered “transparent” panels as well as the creation of “transparent” composite materials with increased thickness, aimed at applications in electrochromic devices. They would be capable of reversible color changes in response to an electric field, which hold significant promise in various fields such as “smart” windows, sensors, displays and military camouflage. The color changes in electrochromic devices generally rely on either oxidation-reduction mechanisms or electrothermal chromic mechanisms. These applications are truly impressive, especially considering that the composites contain insulating materials, such as wood and polymers.

Interestingly, the addition of specific particles tried by Yu et al. (2017), created a composite material, based on wood, suitable for “smart” window applications. It included a dispersion of Cs_xWO_3 nanoparticles, similarly, into pre-polymerized MMA as filling of the nanopores of delignified wood.

In Fu et al. (2018) investigated the creation of “transparent” plywood composite as carrier for bio-sourced synthetic materials with adjustable mechanical and optical properties. They utilized balsa wood (*Ochroma pyramidale*) with a thickness of 0.6–0.8 mm. A 1 wt% NaClO_2 solution in a pH-regulating $(\text{COOH})_2$ solution (pH=4.6) was employed for 6 h at 80 °C to remove lignin, followed by pre-polymerized MMA and QDs (CdSe/ZnS emitting at around 530 nm). The layers were connected in cross-ply composite (0/90/0/-90/0) or quasi-isotropic composite (0/45/90/-45/0) structures, and later polymerization was conducted by heating in an oven at 70 °C for 4 h. Table 2 illustrates the properties measured for these composites.

Li et al. (2018) successfully created highly “transparent” wood composites with a thickness of 1.5 mm and an impressive transmittance of 92%, comparable to the transmittance of pure PMMA at 95% (known as plexiglas, i.e. the artificial glass). The key to this high transmittance, as seen in Fig. 3, lies in altering the wood surfaces through acetylation of end groups. This processing effectively eliminates surface gaps, contributing to the improved transparency of the material, which subsequently

Table 2 Properties resulting from efforts to create “transparent” plywood composites by Fu et al. (2018)

Properties	Units	Single ply composite	Cross-ply composite	Quasi-isotropic composite
Thickness	mm	0.8	3.5	3.5
Cellulose Volume Fraction	%	12	10	10
Young’s modulus (Longitudinal)	GPa	4.3 ± 0.4	4.1 ± 0.3	
Bending strength (longitudinal)	MPa	62.5 ± 2.7	50.1 ± 2.6	
Young’s modulus (transverse)	GPa	2.4 ± 0.2	3.9 ± 0.1	
Bending strength (transverse)	MPa	14.6 ± 1.2	44.9 ± 1.3	
Transmittance	%	90	83	75
Haze	%	50–60	80	80

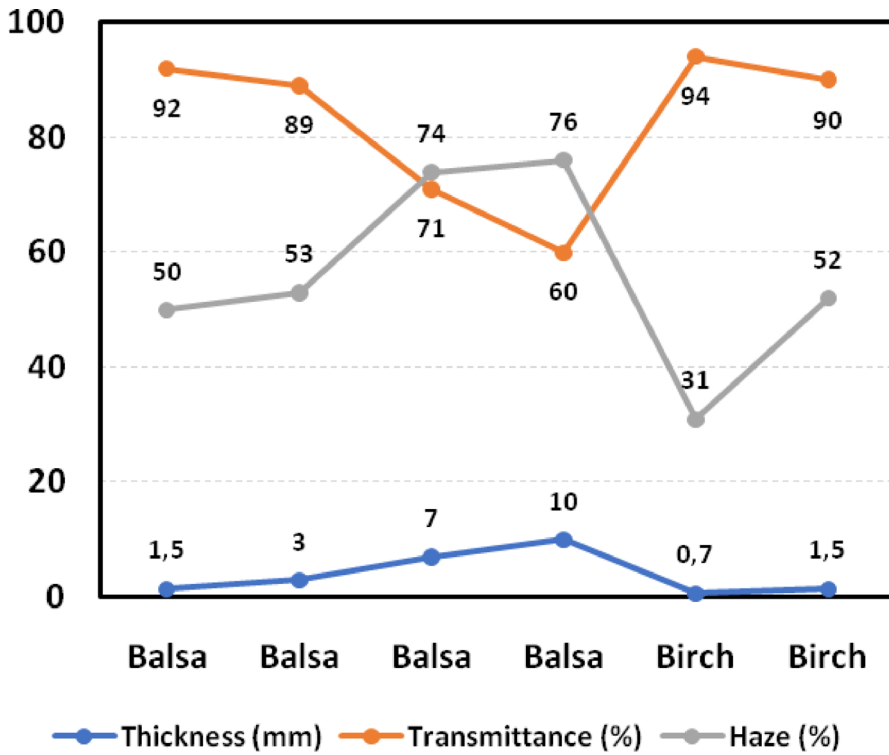


Fig. 3 Optical properties according to the thickness and type of wood studied, according to the research of Li et al. (2018)

finds applications in energy-efficient and “smart” building technologies. Balsa wood was applied in this case, with thicknesses of 1.5, 3, 7, 10 mm, and mahogany wood with thicknesses of 0.7 and 1.5 mm. Delignification took place with 1 wt% NaClO_2 in a pH-regulating $(\text{COOH})_2$ solution at 80 °C. Subsequently, the samples underwent drying in an EtOH and acetone solvent through multiple repetitions. Acetylation was carried out using anhydrous $(\text{COOH})_2$ with pyridine as a catalyst and *N*-methyl-2-pyrrolidone solvent. The acetylation reaction involved temperature range 60 to 120 °C and periods 1 to 12 h. The acetylated wood substrate was further processed with NaClO until it turned white, due to discoloration after acetylation. Before polymer penetration, the wood samples were dehydrated with EtOH and acetone. Pre-polymerized MMA was polymerized at 75 °C for 15 min with 0.3% w/w AIBN as an initiator. The pre-polymerized MMA was then infiltrated into the wood scaffold. The infiltrated composite was placed between two glass slides, wrapped in aluminum foil, and placed in an oven at 70 °C for 4 h for further hardening (i.e. cross-linking inside composite). Another option of post-treatment for increased thickness of the “transparent” wood composites, the pre-polymerized wood template infiltrated with MMA was later sealed in a glass jar and polymerized at 50 °C for 24 h.

The researchers Lang et al. (2018) attempted to apply “transparent” wood composite materials to electrochromic devices capable of customizable color changes. These

impressive “transparent” wood composite materials possess high strength, durability, thermal insulation and transparency. They offer an alternative application to glass for windows with diffuse transparency. These composites exhibit low energy and power consumption, requiring 3 mWh/m² at 2 W/m² to activate. This efficiency is attributed to high coloration efficiency (590 cm²/C) and low voltage (0.8 V). NaClO₂ solution was used for delignification of poplar wood (*Populus*) substrate and MMA followed for the composition of the filler polymer. Additionally, poly(3,4-ethylenedioxythiophene)/tetra-*n*-butylammonium hexafluorophosphate was used as a conductivity enhancer. Various other chemicals and solvents, such as ethylene glycol, ACN, propylene carbonate and toluene, were employed. The reference electrode Ag/Ag⁺ was constructed using a solution of 10 mm AgNO₃/0.5 m TBAPF₆/ACN as the body.

Tang et al. (2018) presented a new and effective method for creating flexible transparent membranes directly from wood, offering improved optical and mechanical properties for use, as transparent protective membranes on mobile phone screens. The transmittance reached 90%, with a conductivity film Ag nanowalls of density of 341 mg/m².

Wang et al. (2018) developed a photochromic “transparent” wood composite with significant progress in the field of “smart” structural materials. This research effort led to a material suitable for various applications, especially in “smart” windows for energy-efficient buildings. The material exhibited excellent thermal insulation, high durability, photochromic properties with the addition of the photochromic material 3',3'-dimethyl-6-nitro-spiro[2*H*-1-benzopyran-2,2'-indoline]-1'-ethanol, allowing the “transparent” composite to undergo reversible color change in response to light exposure. In this case, the material showed a vivid transformation from purple to colorless. It also displayed color-tuning properties with the ability to incorporate dual-chromic paint materials into the photochromic wood substrate, providing additional options for color changes, offering flexibility and adaptability. The photochromic “transparent” composite is created by infiltrating delignified wood with a mixture of the substance mentioned above and pre-polymerized MMA. The developed material promises applications in “smart” windows. These windows could be mechanically robust, energy-efficient and, moreover, capable of serving as photodetectors, allowing users to control window transparency according to lighting conditions Wang et al. (2019). The integration of color-changing properties adds surely an aesthetic dimension to this technology.

In 2019, researchers attempted to optimize the properties of transparent materials based on wood and add additional functionalities to broaden their applications. Efforts were made to create materials for energy storage and application in solar panels or photovoltaics. Photochromic materials were developed, and attempts were made to increase the transparency of these materials. Zhang et al. (2019), created a flexible transparent material derived from wood veneer for alternating current electroluminescent devices, due to its faster and more economical fabrication, compared to transparent membranes based on nanocellulose. Delignification was carried out in a 2%wt NaClO₂ solution at 85°C. The delignified wood veneer was immersed in a wet epoxy resin under vacuum three times to remove air. Coating was done between glass substrate plates and heated to 60°C. A typical three-layer structure was formed: ZnS: Cu embedded in epoxy resin and AgNW/“transparent” wood veneer. AgNWs

were sprayed onto the transparent material, heated and copper tape was soldered for electrical circuitry. ZnS: Cu particles were mixed with epoxy resin and rotationally coated on the bottom electrode. Additional AgNW/"transparent" wood veneer membrane was bonded to the phosphor layer to form the device. Pressure bonding was then performed in a vacuum environment to ensure tight interconnection between layers. The overall transparency of the transparent material achieved a significant value, reaching 86%, surpassing transparent membranes based on cellulose. To make the substrate of the "transparent" wood composite conductive, silver nanowires (in fact AgNWs) were deposited onto it using a spray gun. The resulting conductive "transparent" wood composite membrane exhibited a transparency of 79.5% at a specific surface density of AgNWs, plus a low sheet resistance of $3 \Omega/\text{sq}$. The device exhibited a maximum brightness of 18.36 cd/m^2 at a voltage of 220 V ($f=1 \text{ kHz}$). This performance demonstrates the great potential of the arrangement for signal emission applications.

In Montanari et al. (2019) attempted to develop a material that combines the ability to store thermal energy with transparency, providing effective thermal insulation and the ability to alter optical transparency. Transparent composites, for thermal energy storage systems, can be applied in buildings to save energy. They created "transparent" wood composites for thermal energy storage systems while also adding a, so-called phase-changing, material based on PEG to the "transparent" wood composite. In this study, *Betula pendula* wood with a density of 620 kg/m^3 and dimensions of $2 \text{ cm} \times 2 \text{ cm} \times 0.5 \text{ mm}$ and $2 \text{ cm} \times 2 \text{ cm} \times 1.5 \text{ mm}$ was used for the study, whereas increased sample of $5 \text{ mm} \times 60 \text{ mm} \times 1.5 \text{ mm}$ for mechanical tests. Reagents used were NaClO_2 in $(\text{COOH})_2$ solution (pH 4.6) at 80°C for delignification once again, AIBN and PEG with a MW of 1000 g/mol , for polymer implementation. PEG 1000 was heated to complete dissolution at 45°C . MMA was then added to PEG and stirring continued at 45°C for 10 min. AIBN was then added as an initiator, and stirring continued at 45°C for 10 min. The prepared mixture was infiltrated into the wooden substrate under vacuum at room temperature for about 15 min, repeated three times. The fabricated composites have thicknesses of 0.5 and 1.5 mm and contain a thin layer of PEG/PMMA on their surfaces. The material exhibited high energy storage with a large heat capacity of about 76 J/g , along with the ability to alter optical transparency during the heating process, with an increase in optical transmittance by 6%.

In 2020, efforts were made to create a composite material from wood that would be transparent and, moreover, incorporate a polymer with the quality of flame retardancy, while also having good mechanical properties. For example, Chen et al. (2020), created a "transparent" wood/polyimide composite, which combines the advantages of transparent material and the properties of polyamide in flame retardancy. "Transparent" wood composite has already high transparency, durability, elasticity and mechanical strength, but the inherent flammability has limited its wide applications in electronics. The composite material fabricated was created by impregnating polyimide into delignified wood. The polyimide used is known for its high thermal stability, mechanical strength and flame-spread retardancy. The mechanical strength of the composite is reported to be 42 times higher than that of natural wood. The flame retardant properties make it suitable for applications where fire safety is crucial.

In Fu et al. (2020a, b) developed an innovative approach to produce flexible electronic circuits using materials derived from wood. Specifically, they used Balsa wood with dimensions of $50 \times 50 \times 1 \text{ mm}^3$ (longitudinal $\times$ radial $\times$ tangential). The researchers delignified the samples with 1% NaClO_2 in $(\text{COOH})_2$ solution (pH 4.6) at 80°C for 6 h. They were rinsed with deionized H_2O three times under vacuum and then extracted in a 1 M NaOH solution at room temperature for 3–6 h, followed by three rinses with deionized H_2O . Subsequently, polyethersulfone membranes and release paper were placed between the layers, and pressure was applied with a load of 0.5–5 kg at room temperature, followed by drying for 24–48 h, in order to obtain flexible composite. Then, inks were prepared from lignin-derived carbon fibers. The ink was coated on the wood composite substrate using a coating rod. The described method involves delignification, extraction, ink preparation and printing processes to achieve the final flexible electronic device.

In Yin et al. (2023) prepared a phosphorized “transparent” wood composite. Balsa wood pieces ($30 \text{ mm} \times 30 \text{ mm} \times 2 \text{ mm}$) were immersed in a 5 wt% NaOH solution at 90°C for 2 h. Subsequently, delignification was carried out in a NaClO_2 solution with pH 4.6 at 80°C until they turned white. After delignification, the wood pieces were impregnated with epoxy resin under vacuum and polymerized at room temperature for 72 h to produce “transparent” wood pieces. Blue $g\text{-C}_3\text{N}_4$ was heated to 550°C in a muffle furnace, followed by reflux in concentrated HNO_3 at 80°C for 6 h, while green $g\text{-C}_3\text{N}_4$ was heated to 400°C in a tubular furnace for 40 min. Then, dispersion in epoxy resin was carried out: $g\text{-C}_3\text{N}_4$ (B), $g\text{-C}_3\text{N}_4$ (G) and red phosphorus were separately dispersed in epoxy resin to obtain blue, green and red epoxy resins, which were mixed in a mass ratio of 4:2:1, then a hardening agent was added. This mixture was used to prepare white phosphorized “transparent” wood composites using the same method as for ordinary wood composite preparation. This new material appears transparent under natural light, with phosphor particles uniformly distributed in the epoxy resin. Under UV radiation, $g\text{-C}_3\text{N}_4(\text{W})@\text{TW}$ emits intense white light. $g\text{-C}_3\text{N}_4(\text{W})@\text{TW}$ exhibits high light transmittance (87.17%) and high haze (92.27%) at a wavelength of 550 nm. The thermal conductivity of $g\text{-C}_3\text{N}_4(\text{W})@\text{TW}$ is slightly higher than natural wood but much lower than glass, making it an effective insulating material.

In Chen et al. (2023) created a composite material based on wood called “hydrochromic transparent” wood, which detects changes in humidity by incorporating oxazoline molecules into “transparent” wood composite. One of the key features of “hydrochromic transparent” wood is its ability to change color in response to different humidity levels. This color change is attributed to the reversible opening or closing of oxazoline rings. This property allows the material to exhibit stable color changes under different humidity conditions, detect fingerprints and even display various patterns on its surface when exposed to humidity levels. These functions highlight potential applications of this material in “smart” buildings, intelligent sensors and simulated detectors. The wood used was *Populus cathayana* with dimensions of $20 \times 20 \times 0.50 \text{ mm}^3$ (longitudinal $\times$ tangential $\times$ radial) and a density of 0.386 g/cm . Delignification was carried out in 1 wt% NaClO_2 with an $(\text{COOH})_2$ regulator solution (pH 4.6) at 80°C for 4 to 6 h until completely whitened. The samples were washed three times with deionized H_2O to remove chemical residues. They were

then placed in a mixture of EtOH and acetone (volume ratio 1:1) for 20 min using a vacuum chamber, then stored in acetone before further use. Pre-polymerization with MMA at 75 °C for 15 min with AIBN 0.3% wt. as initiator, and 4 mg H₂O-chromic molecule oxazolidines was added to 10 g of pre-polymerized MMA solution. The mixture was magnetically stirred for 10 min until a uniform dark red color was obtained. The wood samples were impregnated with PMMA/ oxazolidines solutions under −0.85 bar vacuum pressure for 30 min. This process was repeated three times to ensure complete impregnation. The impregnated wood was placed between two glass sheets, wrapped in aluminum foil and hardened in a drying oven at 75 °C for 4 h. “Hydrochromic transparent” wood composite and traditional “transparent” wood composite were separated from the glass after complete polymerization. The tensile strength (9.6 MPa) and modulus of elasticity (1.89 GPa) of hydro-treated composite are nine and six times higher than those of the original wood, respectively. The tensile strength of the original wood (58.4 MPa) is higher than that of the delignified wood (28.95 MPa) due to the remaining pores. Ordinary wood composite exhibits higher tensile strength (93.01 MPa) and modulus of elasticity (5.34 GPa) due to reinforcement with PMMA. Hydro-treated wood composite shows further increases in tensile strength (107.72 MPa) and modulus of elasticity (5.85 GPa) due to improved surface smoothness caused by waterchromic molecule oxazolidines. The biobased “hydrochromic transparent” wood composite with reversible color change under different humidity conditions was successfully developed by impregnating PMMA and hydrochromic molecule oxazolidines into delignified wood. The color of “hydrochromic transparent” wood composites change from dark to light pink with decreasing humidity, due to the chemical structure changes of the oxazoline rings. The hydrogen bond from oxazolidine molecules induces this color change. The homogeneous distribution of H₂O-chromic molecule oxazolidines in “hydrochromic transparent” wood composites improves certainly all its behaviour, including thermal stability, optical and mechanical properties.

In the recent research by Zhou et al. (2024), an effective method was presented for constructing high-quality wood-based composite materials, fully biosourced, providing a sustainable and environmentally friendly alternative to traditional polymers (and polymeric processes as well). They used balsa wood and followed the steps below: delignification of wood to obtain cellulose wood skeleton, oxidation using periodically Na for the introduction of reactive aldehydes, then introduction of gelatin into oxidized wood and physical cross-linking through tannins. They investigated optical properties, mechanical strength, water resistance, UV blocking and bio-degradation, where “transparent” wood film composite and tannic acid/gelatin/film can be biodegraded under natural conditions within 5 months. Microstructural changes where the transformation from oxidized cellulose wood templates to “transparent” wood film composite leads to a compact microstructure attributed to the formation of intra-molecular hemiacetals during oxidation. Color transformation ranged from white (oxidized cellulose wood) to transparent (film) to brown (tannic acid/gelatin/film). Surface characteristic was smooth and flat surface with dense and continuous morphology in tannic acid/gelatin/film. For previous “transparent” wood membranes (without polymer impregnation, thickness: 0.05–0.08 mm) were noted high transparency (70–90%) and high haze (75–90%).

The analysis above demonstrates the variables that play key role to each step of the processes. Times, temperatures, reactants, wood types, polymers and thickness are some of them. Especially wood, which is considered often a “living” material may alter the conditions a researcher plans to apply for certain results to come. For researchers who plan to exercise those delignification techniques and polymer implementation processes, Table 3 summarizes the main objectives and descriptions, advantages and disadvantages of the methods and processes applied, found and retrieved during this review study. We consider those short, collective descriptions, helpful for the experts who are interested in experimenting with “transparent” wood composite manufacture. It is emphasized though that Table 3 contains methods that may eventually not produce that “transparent” wood composite materials.

Table 3 Summary of different methods used to produce “transparent” wood composites using polymers

Production Process	Description	Advantages	Disadvantages
Chemical Delignification & Impregnation:	Removing lignin from wood through chemical treatments (<i>pe.</i> bleaching or acid hydrolysis) to enhance transparency. Then, the wood is impregnated with a transparent polymer, (<i>pe.</i> epoxy or acrylic) to fill the pores and reinforce the structure.	High transparency and strength can be achieved. Control over the degree of delignification and polymer impregnation is allowed.	Chemical treatments may be harsh and require careful handling. Achieving uniform impregnation throughout the wood can be challenging.
Polymer Infiltration & Swelling	Wood is first soaked in a monomer solution that penetrates the cell walls and swells the wood structure. Subsequently, polymerization is initiated within the wood matrix to create a polymer network that imparts transparency.	Results in good transparency and mechanical properties. Allows for the use of a wide range of polymers and monomers	Controlling the polymerization process within the wood matrix to achieve uniformity and prevent cracking or distortion can be difficult.
Nanocellulose Reinforcement:	Involves incorporation of nanocellulose, derived from wood fibers, into a transparent polymer matrix. The nanocellulose serves as reinforcement, improving the mechanical properties of the composite, while maintaining transparency.	Utilizes renewable and biodegradable nanomaterials. Can enhance both transparency and mechanical strength.	Achieving homogeneous dispersion of nanocellulose within the polymer matrix and controlling interactions between nanocellulose and polymer can be challenging.
Layer-by-Layer Assembly	Thin layers of transparent polymers and wood veneers are alternately deposited to create a laminate structure. The polymer layers fill the voids between wood fibers, enhancing transparency and mechanical properties.	Allows for precise control over layer thickness and composition. Can achieve high transparency and customizable properties.	Requires precise deposition techniques/equipment and may result in increased material waste.
Biomimetic Mineralization	Involves infiltrating wood with solutions containing mineral precursors, such as CaCO_3 or silica, followed by controlled mineralization to create a transparent and mineral-reinforced wood composite.	Mimics natural materials and processes. Can enhance both transparency and mechanical properties.	Requires precise control over mineralization process and may result in increased material brittleness.

Table 4 Optical properties achieved by “transparent” wood composites according to the literature, where available

Reference	Wood Species	Sample thickness (mm)	Polymer Type	Delignification Solution	Transparency (%) Haze (%)	Application
Li et al. (2024)	Balsa wood	–	Epoxy resin (EP), PEG, TEP	NaClO_2 , glacial CH_3COOH , 20% H_2O_2	95	Energy-storage, Flame-retardant
Zhou et al. (2024)	Balsa wood	$30 \times 30 \times 1$	Epoxy resin	NaOH , H_3PO_4 , H_2SO_4	79.7–70.3	UV_A & UV_B blocking “Smart” window “Smart” window
Binyaseen et al. (2024)	Basswood	$0.2 \times 25 \times 30$	PMA	H_2O , NaOH (3%), DTPA (0.1%), MgSO_4 (0.1%), NaSiO_4 (3%), H_2O_2 (4%)	–	–
Hu et al. (2024)	Balsawood (longitudinal)	$3 \times 3 \text{ cm}$ $5 \times 5 \text{ cm}$	Aero-Ma-rine epoxy resin (#300)	NaSiO_4 , NaOH , MgSO_4 , diethylenetriaminepenta acetic acid, H_2O_2	55.7–20.3	Enhanced comfort, UV protection
Lü et al. (2024)	Balsa wood	$10.20 \times 10.20 \times 1.04$	PVA with arylboronic acids	NaClO_2 (aq. 5 wt%), pH 4.6 (CH_3COOH)	80–90	Mechanical robust colorful composite
Zhou et al. (2024)	Balsa wood	$30 \times 30 \times 1$	Sodium periodate, Acetic acid	Hydroxylamine hydrochloride, Thymol blue, Tannic acid, Gelatin	27.9–90.3 18.65	“Transparent” composite
Tian et al. (2024)	Balsa wood	$3 \times 3 \times 0.08$	Epoxy resin	NMP solution	–	–
Wu et al. (2023)	Balsa wood	–	–	NaClO_2 /Acetate	70.4–43.5 50.5–65.6	–
Chen et al. (2023)	Balsa wood	$20 \times 20 \times 0.5$	Epoxy resin (E44), PVP, PMMA	NaClO_2 , glacial CH_3COOH , H_2O_2 , EtOH	–	–
Wu et al. (2023)	Poplar wood	–	PVA (MW 75,000, > 99%)	NaClO_2 , 80%	–	–
Du et al. (2023)	Balsa wood	$10 \times 10 \times 1.5$	PMMA (99%)	AQA (30%), NaOH (10%), EtOH, acetone	62–31	–

Table 4 (continued)

Reference	Wood Species	Sample thickness (mm)	Polymer Type	Delignification Solution	Transparency (%) Haze (%)	Application
Zhang et al. (2023)	Poplar wood	60 × 10 × 2.5	PMMA	NaClO ₂ , 80 wt%	–	–
Chu et al. (2022)	Balsa wood	–	PEAG	NaOH-Na ₂ SO ₄	93 98	–
Samanta et al. (2022)	Balsa wood	1.1	MF	Alkaline/H ₂ O ₂	65–92 68–90	–
Jiang et al. (2022)	Rotary-cut basswood and pinewood	0.37		NaClO ₂	~85 ~72	–
Yang et al. (2023)	Balsa wood	0.1–0.7		NaClO ₂	69 73	Hydrophobic wood
Wang et al. (2022)	Balsa wood	2.0	PVP	TEMPO-mediated oxidation of PNIPAM	85.8 ~80	–
Montanari et al. (2021)	Balsa veneers	1.2		Peracetic acid delignification + Succinylation	90 30	–
Van Hai et al. (2022)	Fir wood veneer	0.1		NaClO ₂ delignification	80 30–60	–
Li et al. (2020)	Balsa wood	1.5		NaClO ₂ delignification	72–80	–
Wu et al. (2019)	Poplar wood	1.5 (3 layers), 2.5 (5 layers)		NaClO ₂ delignification	~1.2 (3 layers) - ~0.45 (5 layers)	–
Höglund et al. (2021)	Balsa wood	1.0		NaClO ₂ (1 wt%) + acetate buffer solution, pH=4.6 + 1 M NaOH immersion	81 –	–
Li et al. (2019)	Basswood/Pinewood	50/100 × 50 × 5	Epoxy resin	Boiling H ₂ O ₂ aqueous (30 wt%)	87 90	–
Katunský et al. (2018)	Balsa wood	0.7–3.7	MMA	NaClO ₂	40–90 90	–

Table 4 (continued)

Reference	Wood Species	Sample thickness (mm)	Polymer Type	Delignification Solution	Transparency (%) Haze (%)	Application
Fu et al. (2018)	Balsa wood	3.5 (5 layers)	—	NaClO ₂	83	—
Li et al. (2018)	Balsa	1.5	PMMA	NaClO ₂ /Acetate	83 70	—
Bi et al. (2018)	Balsa	200 × 200 × 1	—	PhCl/Oxalic Acid DES, Microwave heating	92 85	White Light-Emitting Diodes
Li, Fu et al., (2017)	Balsa	1.5	PMMA	NaClO ₂ /Acetate, Alkaline/ H ₂ O ₂	83–86 68–75	Luminescent “Transparent” Wood
Jia et al. (2017)	Basswood	0.8	—	5 wt% NaClO solution	~90 ~90	—
Yaddanapudi et al. (2017)	Coniferous (Cellulose)	0.13	—	Swelled, dispersed, infiltrated, hot-pressed pulp	~86 ~90	“Smart” building applications
Yaddanapudi et al. (2017)	<i>Fagus sylvatica</i>	0.1	PMMA	NaClO ₂ (5 wt%) in acetate buffer	70–15 18–49	“Transparent” wood
Yu et al. (2017)	Beech	60 × 30 × 5	PMMA	—	86 90	—
Li et al. (2016)	Balsa wood	0.7–3.7	PMMA	NaClO ₂ /Acetate	34–90 48–80 86–90	—
Gan et al. (2017b)	Cathay poplar (<i>Populus cathayana</i>)	20 × 10 × 0.5	—	—	—	—
Zhu et al. (2016a, b)	Basswood	3	Epoxy resin EP	NaOH–Na ₂ SO ₄ , H ₂ O ₂	~80–90	—
Fang and Zhu (2014)	Southern yellow pine	—	—	TEMPO-oxidized system	~96 ~60	—

Table 5 Mechanical properties achieved by “transparent” wood composites according to literature, where available

Reference	Wood Species	Sample Thickness (mm)	Polymer Type	Delignification Treatment	Young's Modulus (E) (GPa)	Tensile Strength (σ) (MPa)	Elongation at Break (ϵ) (%)	Application
Li et al. (2024)	Balsa	–	Epoxy resin, PEG, TEP	NaClO ₂ , glacial CH ₃ COOH, 20 wt% H ₂ O ₂	NW: 0.84, DW: 0.48, EP: 45.89, TW/PEG/TEP: 1.79	NW: 2.78, DW: 8.03, EP: 25.97, TW/PEG/TEP: 14.21	–	Flame-retardant, energy-storage
Zhou et al. (2024)	Balsa	30 × 30 × 1	Epoxy resin	NaOH, H ₂ PO ₄ , H ₂ SO ₄	–	LW-TW: 16.25–19.39	LW-TW: 2.86–5.35%	“Smart” window
Binyaseen et al. (2024)	Basswood	0.2 × 25 × 30	PMA	NaOH, DTPA, MgSO ₄ , Na ₂ SiO ₃ , H ₂ O ₂	–	–	–	“Smart” window
Hu et al. (2024)	Balsa (longitudinal)	3 × 3 cm 5 × 5 cm	Aero-Marine epoxy resin	Na ₂ SiO ₃ , NaOH, MgSO ₄ , DTPA, 8 wt% H ₂ O ₂	NW: 23.18, EMTW: 59.98, BARETW: 59.61	NW: 1.5, EMTW: 3, BARETW: 3.4	–	UV-blocking properties
Lü et al. (2024)	Balsa	10.2 × 10.2 × 1.04	PVA (with arylboronic acids)	NaClO ₂ (5 wt%), CH ₃ COOH	–	W-3Bp-B: 154, W-Ph9-B: 141	–	Mechanically robust, colorful composite
Tian et al. (2024)	Balsa	3 × 3 × 0.08 cm	Epoxy resin	NMP solution	11.19 (NW), 11.16 (PW), 0.65 (PTW)	NW: 0.122 GPa, PW: 0.115 GPa, PTW: 0.98 GPa	NW: 1.09%, PW: 1.03%, PTW: 149.69%	Flexible, photore sponsive, “transparent” composite
Wu et al. (2023)	Balsa	0.48	–	NaClO ₂ /Acetate	99.5	–	–	Hydrophobic wood
Zhang et al. (2023)	Poplar	10 × 10 × 10, 60 × 10 × 2.5	PMMA	NaClO ₂ (80 wt%)	–	–	–	“Smart” window

Table 5 (continued)

Reference	Wood Species	Sample Thickness (mm)	Polymer Type	Delignification Treatment	Young's Modulus (E) (GPa)	Tensile Strength (σ) (MPa)	Elongation at Break (ϵ) (%)	Application
Chu et al. (2022)	Balsa	Not specified	PEAG, Epoxy resin	NaOH+Na ₂ SO ₄ , H ₂ O ₂	2.2	153.6	—	“Trans-parent” composite
Jiang et al. (2022)	Basswood, Pinewood	0.37, ~45 μ m (Sugarcane bagasse fiber)	—	NaClO ₂ , H ₂ O ₂ + UV	165.3 MPa (Basswood, Pinewood), 38 MPa (Sugarcane)	—	—	Hydrophobic and UV-resistant “transparent” composite
Montanari et al. (2021)	Balsa veneers	1.2	—	Peracetic acid delignification + Succinylation	17 GPa	174	—	Structural applications
Wang et al. (2021)	Balsa	2	—	NaClO ₂	~25 MPa (3 layers), ~45 MPa (5 layers)	—	—	“Transpar-ent” wood composites
Montanari et al. (2020)	Birch	0.7–1.2	—	PAA-NaOH	12.6–17.3	146.6–173.3	1.1–1.2	—
Xia et al. (2021a)	Balsa	1.0 & 1.5	Epoxy resin EP	UV-H ₂ O ₂ -NaOH	—	46.2 & 31.4	3.4&7.4	—
Mi et al. (2020a, b)	Balsa	0.8	PVA	NaClO & none	3.85 & 3.1	143& 67	4.2 & 3.7	—
Höglund et al. (2020)	Balsa	1.2	—	NaClO ₂ /Acetate Alkaline/H ₂ O ₂	3.2 3.4	53.7 59	2.1 2.5	—
Lu et al. (2020)	Poplar	—	—	NaClO ₂	1.0	—	—	—
Khakalo et al. (2020)	Birch wood plies	—	—	NaClO ₂ + ionic liquid	—	—	—	—
Jiang et al. (2020)	Raw flax fibers (<i>Linum usitatissimum</i>)	—	—	NaOH + NaSO ₃ + H ₂ O ₂	1.5	—	—	—

Table 5 (continued)

Reference	Wood Species	Sample Thickness (mm)	Polymer Type	Delignification Treatment	Young's Modulus (E) (GPa)	Tensile Strength (σ) (MPa)	Elongation at Break (ϵ) (%)	Application
Mi et al. (2020a, b)	Beech wood	–	–	NaClO ₂ (5 wt%) + acetate buffer solution (pH = 4.6)	–	0.1–0.7	–	–
Fu, Chen et al. (2020)	Wheat straw	–	–	NaOH (10 wt%) + Na ₂ SO ₃ (5 wt%) + H ₂ O ₂ (30 wt%)	–	3 mm (30 wt%)	–	–
Li et al. (2019)	Basswood/Pinewood	50 or 100 × 50 × 5	Epoxy resin	Boiling H ₂ O ₂ (aq. 30 wt%)	–	–	–	–
Li et al. (2018)	Balsa	2.0	PMMA	PAA-NaOH	2.9–4.8	11.4–41.4	–	–
Montanari et al. (2019)	Birch	0.5	PMMA, PEG	NaClO ₂ /Acetate	–	14.9	70.5	–
Khakalo et al. (2019)	Birch	–	–	NaClO ₂	430	–	–	–
Jia et al. (2019)	Basswood	0.7	Epoxy resin EP	NaClO NaClO ₂ /Acetate	– –	33.29–43.39 –	– –	– –
Qin et al. (2018)	Balsa	2.0	Epoxy resin EP	NaClO ₂ /Acetate	69.13–75.12	6.06–8.17	–	–
Katunský et al. (2018)	Balsa	–	MMA	NaClO ₂ Boiling NaOH-Na ₂ SO ₃ -H ₂ O ₂	2.05 ± 0.13 1.22	90.1 ± 10 23.38	– –	“Transparent” wood
Fu et al. (2018)	Balsa	3.5 (5 layers)	–	NaClO ₂	50 (5 layers)	–	–	–
Li et al. (2018)	Balsa	1.5	PMMA	NaClO ₂ /Acetate	4	78.9	2.2	–

Table 5 (continued)

Reference	Wood Species	Sample Thickness (mm)	Polymer Type	Delignification Treatment	Young's Modulus (E) (GPa)	Tensile Strength (σ) (MPa)	Elongation at Break (ϵ) (%)	Application
Bi et al. (2018)	Balsa	200 × 200 × 1	—	Deep Eutectic Solvent + Microwave-Assisted Heating	—	Fracture strength: 27.38 MPa (CDs/ <i>R</i> -TW composite)	2.2% (CDs/ <i>R</i> -TW)	White Light-Emitting Diodes
Li, Fu et al. (2017)	Balsa	1.5	PMMA	NaClO ₂ /Acetate/Alkaline/H ₂ O ₂	100.7	2.18	—	—
Jia et al. (2017)	Basswood	0.8	—	5 wt% NaClO solution	—	—	—	—
Yaddanapudi et al. (2017)	Cellulose fibers of Coniferous wood	0.13	—	Pulp was swelled, dispersed, infiltrated, hot-pressed	—	140	—	—
Li et al. (2016)	Balsa	0.7–1.2	PMMA	NaClO ₂ /Acetate	2.05–3.59	38–90.1	2.2–3.8	—
Zhu et al. (2016a, b)	Basswood	3.0	Epoxy resin EP	NaOH-Na ₂ SO ₄ +H ₂ O ₂	1.22	23.38	—	—
Fang and Zhu (2014)	Southern yellow pine pulp	—	—	TEMPO-oxidized system	—	~105	—	—
Fink et al. (1992)	European Ash/Leafed Lime/Champion Oak	—	Some Monomeric mixture	NaOCl	—	—	—	“Transparent” wood

Finally, since this is a technical review, detailed, technical data found and collected from research papers are placed in Tables 4 and 5, regarding optical properties (since named “transparent”) and mechanical properties (as anticipated for most “composites”), correspondingly. Readers should keep in mind that substrates are of natural origin, not synthetic, thus variations are sure regarding woods. Additionally, in experimental level, the apparatus available for laboratory manufacture plays a decisive role as well, for the actual reinforcement of a material or the isotropic distribution of its physicochemical properties.

Discussion

The extent parathesis of the results of the research publications chosen, has illustrated clearly the factors involving until nowadays in the manufacturing of “transparent” wood composites, and the reasons why those factors affect it (Tables 3, 4 and 5). On the other hand, the production of “transparent” wood composites impregnated with polymers is an innovative approach for industrial manufactures. Wood is already a natural material, found in abundance, even as side-product of the major wood-processing industries. Most polymers are of petrochemical origin, thus non-renewable source. “Transparent” wood composites crafted with up-cycled polymers though *versus* conventional techniques and ordinary polymers involve primary evaluation of various aspects; functionality, economic viability, industrial availability and environmental impact. It is given that traditional methods often yield superior strength and resilience by impregnating wood with transparent polymers, bolstering its mechanical attributes. Recycled polymers may exhibit comparable characteristics, contingent upon the particular polymer utilized and fabrication method. As for clarity and transparency traditional methods might deliver clearer outcomes, owing to the uniform resin saturation. Recycled polymers could marginally compromise clarity contingent on the recycled material’s quality and manufacturing process intricacies.

Recyclable polymers (not always same as recycled polymers actually) generally offer enhanced cost efficiency, as they harness waste materials, promote circular economy, seem a more environmentally friendly process curbing the necessity for virgin resources. Incorporation of recyclable polymers bolsters sustainability by diverting waste from landfills and diminishing virgin material demands. This is one of the reasons why their incorporation would find little market place, their economics do not favor them. Traditional methods might rely on virgin resins or chemicals, heightening resource utilization. Recycled polymers minimize carbon footprint regarding the sources’ elimination *versus* traditional methods, due to less energy-intensive processes and reduced reliance on fossil fuels. Certainly, the purification processes for inducing a post-consuming polymer into a new application, are not negligible. Furthermore, all polymers are not recyclable in chemical level; PMMA may yield back MMA monomer because of the unzipping chain reactions, but epoxy resins (and in general all cross-linked polymers) cannot give monomers or oligomers due to strong networking and all kinds of degradation reactions.

In essence, while traditional methods might tout superior performance and clarity, “transparent” wood composites crafted with recycled polymers could offer height-

ened cost efficiency and sustainability courtesy of repurposed materials or potentially diminished manufacturing overheads. Selection between the two hinges upon specific application requisites, financial constraints and sustainability aspirations or enterprises' marketing goals. Future research directions and potential improvements in the field of treated wood composites should focus to enhanced clarity and transparency, in the presence of various chemical agents. To be more specific, it is crucial for novel processing techniques or additives to be investigated so the clarity and transparency of "transparent" composites gets improved, making it more suitable for applications requiring high optical quality in certain thicknesses.

Another field that could further improve is mechanical properties. New methods may be explored to further strengthen the mechanical properties of "transparent" wood composites, such as increasing tensile strength and reducing brittleness in their little widths, to expand their range of applications. The dependence of mechanical properties on the geometry and direction of load is crucial and comparisons are hard to make (Table 4).

Additionally, in order for them to be marketed it is useful to be produced in large scale (Xia et al. 2021b) by developing scalable manufacturing processes for wood composites to enable mass production and commercialization, potentially reducing costs and making it more accessible for widespread use. The raw materials costs and energy costs are decisive in this sector, taken for given that infrastructures already exist.

Future research options seem to be functionalities such as UV-resistance, flame retardancy or "self-healing" properties, humidity response and temperature response, all tailored to specific application requirements. This is why those composites with distinguished chemical tiny ingredients, are often called "smart" commercially (Liu et al. 2020, 2022). Moreover, specific trends would appear to be the investigation of eco-friendly and biodegradable additives or coatings in order to enhance the sustainability of "transparent" wood commercial products, ensuring minimal environmental impact throughout their lifecycle. The harder the coatings are for increased qualities over weathering, the harder the biodegradation to occur.

Exploring the integration of "transparent" wood composites with emerging technologies, such as nanotechnology or additive manufacturing (e.g. 3D-printing) to create innovative, multifunctional structures with unique properties, would be of great research interest. Advancing research on the structural design and engineering applications of "transparent" wood composites, including its use in architectural design, lightweight construction and renewable energy systems, has already given results. Furthermore, conducting long-term durability studies to assess the ageing properties and lifespan of those wood materials, ensuring they meet industry standards for longevity and performance, is essential for answering consumers' queries. By addressing these research directions and implementing potential improvements, the field of "transparent" wood composites manufacture could start to evolve massively, leading to the development of innovative materials with a wide range of applications and benefits for society and the environment, as final goal.

Conclusion

This review study searched and chose, based on certain criteria, the scientific publications regarding the innovative manufacturing of “transparent” wood composite materials of great input, for various industrial new-generation applications. The selected papers were *meta*-analysed, based on general interesting data or details regarding environmental issues. Most importantly, exact conditions and reagents regarding wood type, polymer type, delignification and impregnation processes were reviewed. Crucial improved properties of “transparent” wood composites were discussed, such as optical transparency, mechanical strength and thermal stability, and finally, authors commented on potential future actions or goals. The objectives set for this review paper were served and hope readers find it helpful after all, for getting familiar with the technical demands the manufacturing of “transparent” wood composites present.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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